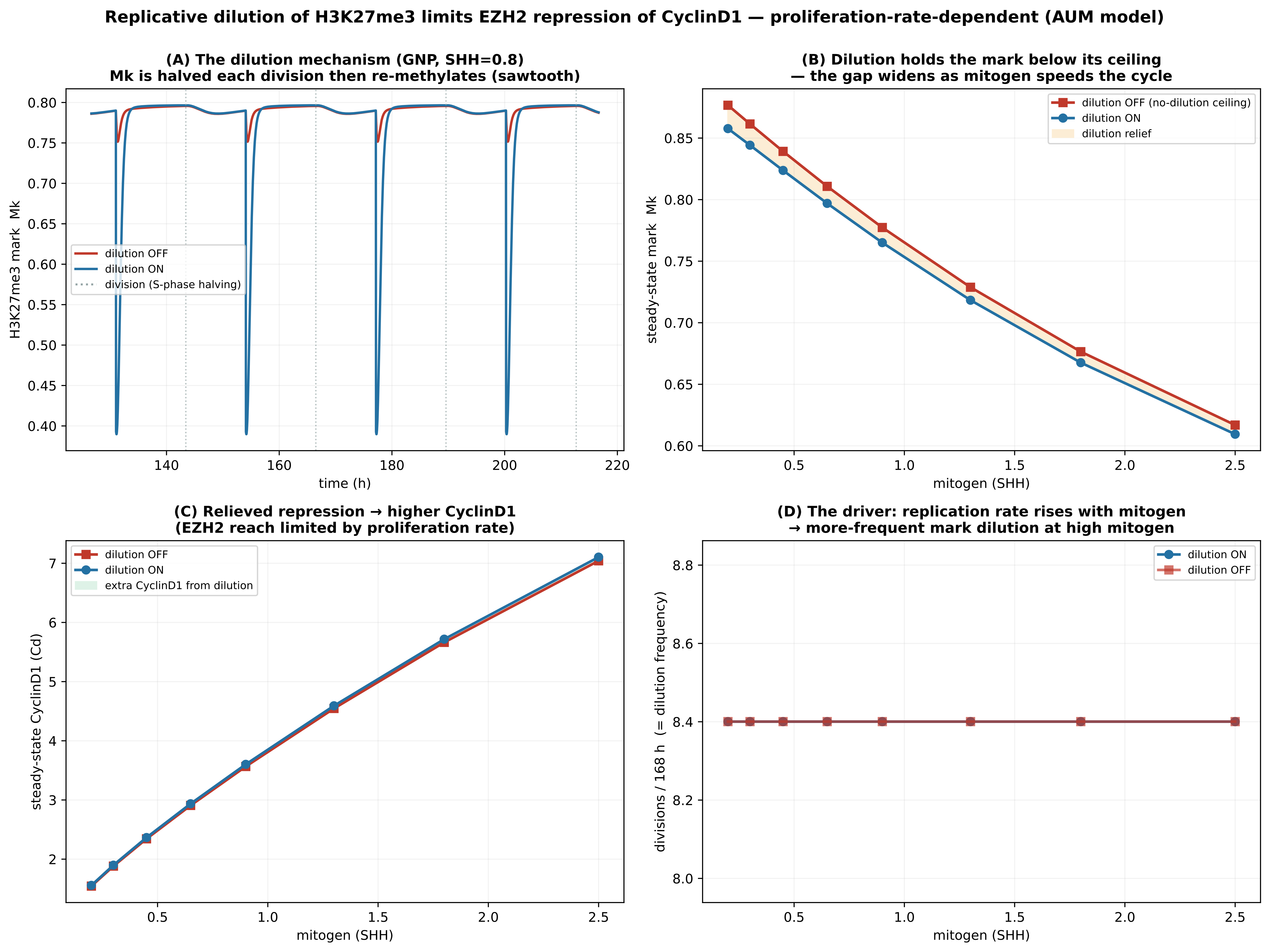

Replicative dilution of H3K27me3 limits EZH2 repression of CyclinD1 — proliferation-rate-dependent (AUM model)
(A) The dilution mechanism (GNP, SHH=0.8)
Mk is halved each division then re-methylates (sawtooth)
H3K27me3 mark Mk
time (h)
dilution OFF
dilution ON
division (S-phase halving)
(B) Dilution holds the mark below its ceiling
— the gap widens as mitogen speeds the cycle
steady-state mark Mk
mitogen (SHH)
dilution OFF (no-dilution ceiling)
dilution ON
dilution relief
(C) Relieved repression → higher CyclinD1
(EZH2 reach limited by proliferation rate)
steady-state CyclinD1 (Cd)
mitogen (SHH)
dilution OFF
dilution ON
extra CyclinD1 from dilution
(D) The driver: replication rate rises with mitogen
→ more-frequent mark dilution at high mitogen
divisions / 168 h (= dilution frequency)
mitogen (SHH)
dilution ON
dilution OFF